

Relativistic Momentum Density, Four-Force Density and Four-Acceleration in Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity

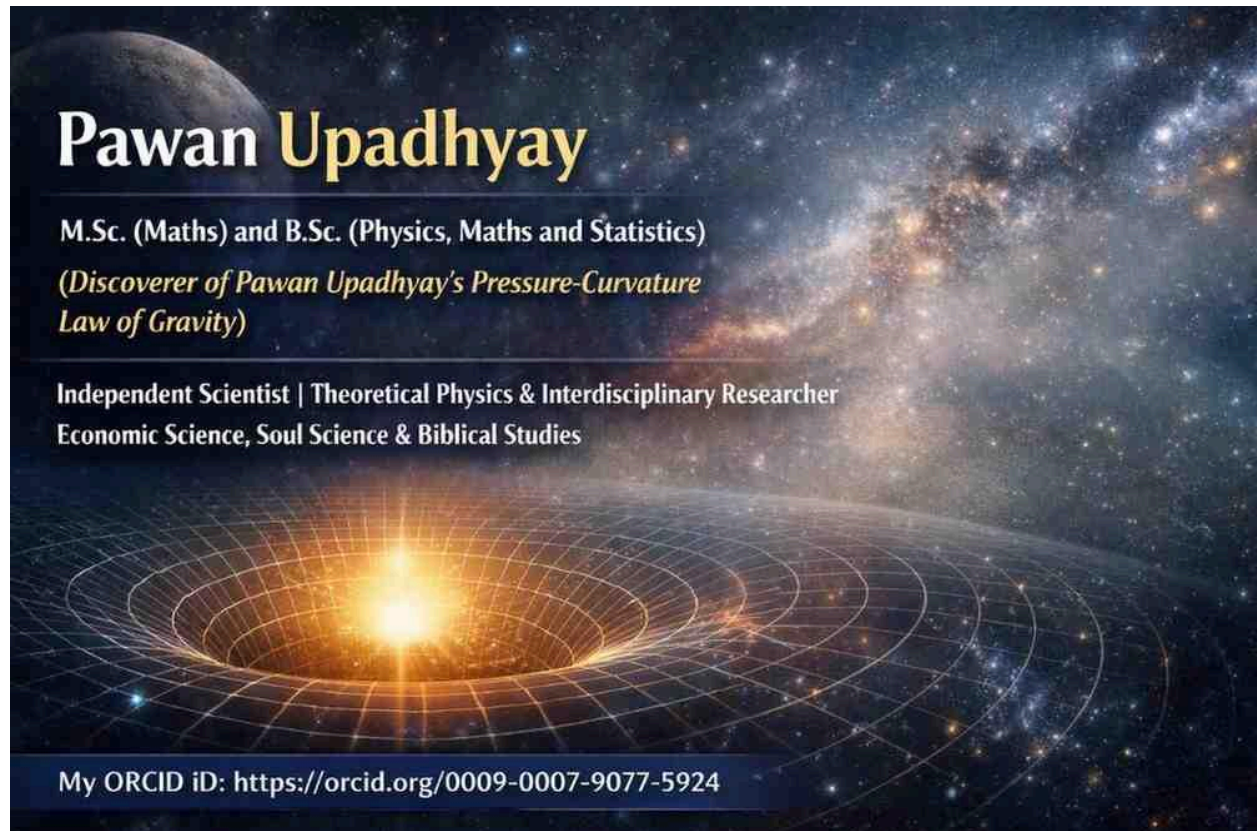
Author: Pawan Upadhyay

Affiliation: Independent Researcher

Email: pawanupadhyay28@hotmail.com

Field: Theoretical Physics, Relativistic Continuum Mechanics, Gravitational Field Theory

ORCID iD: <https://orcid.org/0009-0007-9077-5924>



Abstract

This paper presents a relativistic continuum formulation of four-momentum density, four-force density, and four-acceleration within Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity. The framework interprets gravity as a pressure–curvature-driven spacetime phenomenon in which gravitational pressure gradients generate relativistic force density, spacetime acceleration, and momentum-density flow.

Beginning from relativistic stress–energy conservation, the theory derives a relativistic acceleration equation through projection perpendicular to the four-velocity field. The framework then connects four-force density, four-velocity, four-acceleration, and four-momentum density into a unified continuum gravitational structure.

The central PPC equations are:

$$(E_d + P_g)a^\nu + \nabla^\nu P_g = 0$$

and

$$\frac{dP_d^\mu}{d\tau} = -\nabla^\mu P_g$$

where:

- E_d is relativistic energy density,
- P_g is gravitational pressure density,
- a^ν is four-acceleration,
- P_d^μ is four-momentum density,
- τ is proper time.

The theory attempts to unify:

- pressure density,
- force density,
- relativistic acceleration,

- spacetime flow,
- and momentum-density transport

within a single relativistic continuum gravitational framework.

1. Introduction

Relativistic physics describes matter and energy using spacetime four-vectors and tensor equations. In relativistic continuum systems, local conservation of energy and momentum is expressed through the stress–energy tensor.

Pawan Upadhyay’s Pressure–Curvature (PPC) Law of Gravity interprets gravity as a pressure-gradient-driven spacetime curvature phenomenon arising from distributed gravitational pressure-density structures.

Within this framework:

- pressure gradients generate relativistic force density,
- force density produces spacetime acceleration,
- acceleration generates relativistic momentum-density flow.

The theory extends relativistic momentum concepts into continuum gravitational dynamics.

2. Relativistic Four-Momentum

The relativistic four-momentum vector is:

$$P^\mu = \left(\frac{E}{c}, p_x, p_y, p_z \right)$$

where:

- E/c is the temporal momentum component,
- p_x, p_y, p_z are spatial momentum components.

The quantity E/c possesses momentum dimensions and represents temporal momentum flow in spacetime geometry.

For photons:

$$E = pc$$

thus:

$$\frac{E}{c} = p$$

demonstrating the relativistic relation
between energy and momentum.

3. Relativistic Four-Momentum Density

For distributed relativistic continua, dividing four-momentum by volume gives four-momentum density.

Energy density is:

$$E_d = \frac{E}{V}$$

The four-momentum density vector becomes:

$$P_d^\mu = \left(\frac{E_d}{c}, p_{dx}, p_{dy}, p_{dz} \right)$$

where:

- p_{dx}, p_{dy}, p_{dz} are spatial momentum-density components.

This formulation represents:

- spacetime energy-flow density,
 - relativistic momentum transport density,
 - continuum spacetime motion.
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4. Four-Velocity and Momentum Density

The relativistic four-velocity is written as:

$$U^\mu = (\gamma c, \gamma v_x, \gamma v_y, \gamma v_z)$$

or equivalently:

$$u^\mu = (\gamma c, \gamma v_x, \gamma v_y, \gamma v_z)$$

In this paper:

- U^μ and u^μ represent the same relativistic four-velocity quantity,
- the difference is only notational convention.

Uppercase U^μ is often used for particle four-velocity, while lowercase u^μ is commonly used in relativistic fluid dynamics and continuum mechanics.

The four-momentum density relation is:

$$P_d^\mu = \rho U^\mu$$

or equivalently:

$$P_d^\mu = \rho u^\mu$$

where:

- ρ is relativistic mass density.

Expanded form:

$$P_d^\mu = (\gamma\rho c, \gamma\rho v_x, \gamma\rho v_y, \gamma\rho v_z)$$

Thus relativistic spacetime motion generates momentum-density flow.

5. Local Stress–Energy Conservation

In relativistic continuum mechanics and General Relativity, local conservation of energy and momentum is expressed through:

$$\nabla_{\mu} T^{\mu\nu} = 0$$

This equation represents local conservation of:

- energy,
- momentum,
- relativistic fluid motion.

Projecting this equation perpendicular to the four-velocity gives the relativistic momentum conservation equation.

PPC Stress–Energy Tensor

The PPC relativistic stress–energy tensor is defined as:

$$T^{\mu\nu} = (E_d + P_g)U^\mu U^\nu + P_g g^{\mu\nu}$$

where:

- E_d represents relativistic energy density,
- P_g represents gravitational pressure density,
- U^μ represents relativistic four-velocity,
- $g^{\mu\nu}$ represents the spacetime metric tensor.

The tensor satisfies the local conservation equation:

$$\nabla_{\mu} T^{\mu\nu} = 0$$

Projecting the conservation equation perpendicular to the four-velocity yields the PPC relativistic acceleration equation:

$$(E_d + P_g) a^{\nu} + \nabla^{\nu} P_g = 0$$

This equation describes relativistic acceleration generated by gravitational pressure gradients within spacetime continua.

6. Four-Acceleration in PPC Theory

The relativistic four-acceleration field is:

$$a^\nu = u^\mu \nabla_\mu u^\nu$$

or equivalently:

$$a^\nu = U^\mu \nabla_\mu U^\nu$$

where:

- u^μ and U^μ represent the same four-velocity field,
- ∇_μ is the covariant derivative operator.

This equation describes the covariant rate of change of spacetime velocity along the relativistic flow.

7. PPC Relativistic Acceleration Equation

Projecting the conservation equation perpendicular to the four-velocity gives the PPC relativistic acceleration equation:

$$(E_d + P_g)a^\nu + \nabla^\nu P_g = 0$$

where:

- E_d is relativistic energy density,
- P_g is gravitational pressure density,
- a^ν is four-acceleration.

Rearranging gives:

$$a^\nu = -\frac{\nabla^\nu P_g}{E_d + P_g}$$

8. Physical Interpretation of the PPC Acceleration Equation

The PPC acceleration equation implies:

- covariant variation in gravitational pressure generates acceleration,
- $(E_d + P_g)$ behaves like effective relativistic inertial density,
- pressure contributes to relativistic inertia similarly to energy density.

Thus:

- pressure resists acceleration,
- pressure gradients drive spacetime motion,
- relativistic dynamics emerge from pressure-curvature interactions.

This extends classical inertial concepts into relativistic continuum geometry.

9. Four-Force Density in PPC Theory

The PPC relativistic four-force density equation is:

$$F^\mu = -\nabla^\mu P_g$$

where:

- F^μ is four-force density,

The negative sign indicates that force density acts toward decreasing gravitational pressure.

This equation interprets gravity as a pressure-gradient-driven spacetime force-density phenomenon.

10. Momentum Density Evolution Equation

The relativistic evolution equation for four-momentum density is:

$$F^\mu = \frac{dP_d^\mu}{d\tau}$$

where:

- τ denotes relativistic proper time.

Combining with the PPC force-density equation gives:

$$\frac{dP_d^\mu}{d\tau} = -\nabla^\mu P_g$$

This equation states:

the relativistic rate of change of four-momentum density equals the negative covariant gravitational pressure gradient.

11. PPC Dynamical Structure

The PPC spacetime continuum structure may be expressed as:

$$P_g \rightarrow \nabla^\mu P_g \rightarrow F^\mu \rightarrow a^\nu \rightarrow u^\nu \rightarrow P_d^\nu$$

where:

- P_g represents gravitational pressure density,
- $\nabla^\mu P_g$ represents the covariant gravitational pressure gradient,
- F^μ represents four-force density,
- a^ν represents relativistic four-acceleration,
- u^ν represents relativistic four-velocity,
- P_d^ν represents relativistic four-momentum density.

The relativistic four-acceleration field is:

$$a^\nu = u^\mu \nabla_\mu u^\nu$$

and the momentum-density evolution equation is:

$$\frac{dP_d^\nu}{d\tau} = -\nabla^\nu P_g$$

Physical interpretation of the PPC dynamical structure:

1. gravitational pressure density exists throughout spacetime,
2. covariant pressure gradients form within the gravitational field,
3. pressure gradients generate relativistic four-force density,
4. four-force density produces relativistic four-acceleration,
5. four-acceleration changes the spacetime four-velocity field,
6. four-velocity generates relativistic four-momentum density flow.

Within PPC Theory, gravity is therefore interpreted as a relativistic pressure-curvature-driven spacetime flow phenomenon generated by distributed gravitational pressure-density gradients.

OR

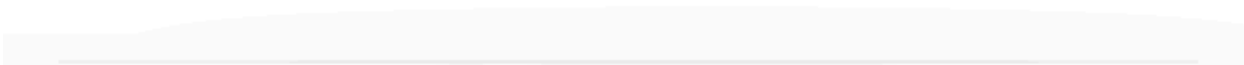
In Short Form,

$$P_g \rightarrow \nabla^\mu P_g \rightarrow F^\mu \rightarrow \frac{dP_d^\mu}{d\tau} \rightarrow U^\mu \rightarrow P_d^\mu$$

or equivalently:

$$P_g \rightarrow \nabla^\mu P_g \rightarrow F^\mu \rightarrow \frac{dP_d^\mu}{d\tau} \rightarrow u^\mu \rightarrow P_d^\mu$$

Physical interpretation:

1. gravitational pressure density exists,
 2. pressure gradients form,
 3. gradients generate four-force density,
 4. force density evolves four-momentum density,
 5. relativistic motion generates four-velocity,
 6. four-velocity produces spacetime momentum-density flow.
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12. Relation to Relativistic Continuum Mechanics

The PPC framework naturally connects with:

- relativistic hydrodynamics,
- relativistic fluid mechanics,
- stress–energy tensor systems,
- continuum momentum transport,
- relativistic energy-flow theory.

The theory attempts to interpret gravity as a relativistic pressure-flow phenomenon embedded within spacetime continua.

13. Geometric Interpretation

Within PPC Theory:

- P_g represents gravitational pressure structure,
- $\nabla^\mu P_g$ represents curvature-driving gradients,
- F^μ represents spacetime force density,
- a^ν represents relativistic spacetime acceleration,
- P_d^μ represents momentum-density flow.

Gravity is therefore interpreted as:

- a pressure-curvature spacetime flow phenomenon,
- generated by distributed gravitational pressure-density gradients.

14. Conclusion

This paper developed a relativistic continuum formulation of:

- four-momentum density,
- four-force density,
- and four-acceleration

within Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity.

The central PPC equations:

$$(E_d + P_g)a^\nu + \nabla^\nu P_g = 0$$

and

$$\frac{dP_d^\mu}{d\tau} = -\nabla^\mu P_g$$

interpret gravity as a relativistic pressure-gradient-driven momentum-density phenomenon.

The framework attempts to unify:

- pressure density,
- force density,
- relativistic acceleration,
- spacetime motion,
- and momentum-density transport

within a single relativistic continuum gravitational theory.

15. In Weak Field Limit

Weak-Field Limit in PPC Theory

In the weak-field and low-velocity limit, the relativistic PPC framework reduces to classical momentum and momentum-density dynamics.

The classical force equation becomes:

$$f = \frac{dp}{dt}$$

where:

- f is classical force,
- p is classical momentum,
- t is coordinate time.

For distributed continuum systems, the force-density equation becomes:

$$F = \frac{dp_d}{dt}$$

where:

- F is classical force density,
- p_d is classical momentum density.

Thus:

- lowercase f represents ordinary force acting on localized matter,
- uppercase F represents force density acting on distributed continuum matter systems.

The clearer and standard notation is:

$$F = \frac{1}{dV} \frac{dp}{dt}$$

or equivalently:

$$F = \frac{d}{dV} \left(\frac{dp}{dt} \right)$$

where:

- F = force density,
- dV = differential volume element,
- p = momentum.

Since momentum density is:

$$p_d = \frac{dp}{dV}$$

then:

$$F = \frac{dp_d}{dt}$$

- force density = time rate of change of momentum density,
- or equivalently force per unit volume.

In the strong-field relativistic PPC framework, these equations generalize into covariant spacetime form:

$$F^\mu = -\nabla^\mu P_g$$

and

$$\frac{dP_d^\mu}{d\tau} = -\nabla^\mu P_g$$

where:

- F^μ is relativistic four-force density,
- P_d^μ is relativistic four-momentum density,
- P_g is gravitational pressure density,
- τ is relativistic proper time.

Therefore:

- weak-field PPC dynamics reduce to classical force and force-density equations,
- strong-field PPC dynamics become relativistic pressure-curvature continuum equations.

Important Clarification:

$$F^\nu = -\nabla^\nu P_g$$

then:

$$F^\nu \rightarrow a^\nu$$

then:

$$a^\nu \rightarrow U^\nu$$

then:

$$P_d^\nu = \rho U^\nu$$

So the physically clearer chain is:

$$P_g \rightarrow \nabla^\nu P_g \rightarrow F^\nu \rightarrow a^\nu \rightarrow U^\nu \rightarrow P_d^\nu$$

- pressure gradient generates force density,
- force density generates acceleration,
- acceleration changes four-velocity,
- four-velocity determines four-momentum density.

That sequence aligns better with:

- relativistic continuum mechanics,
- relativistic fluid dynamics,
- and covariant field equations.

In relativistic tensor notation, the index symbol itself is arbitrary.

So:

$$P_d^\nu$$

and

$$P_d^\mu$$

represent the same type of four-vector quantity.

$$F^\nu = \frac{dP_d^\nu}{d\tau}$$

or:

$$F^\mu = \frac{dP_d^\mu}{d\tau}$$

Both are mathematically equivalent.

Similarly:

$$F^\mu = -\nabla^\mu P_g$$

or:

$$F^\nu = -\nabla^\nu P_g$$

are equivalent representations.

- u^μ and u^ν are the same type of four-vector quantity,
- the Greek index changes depending on the equation structure.